

## ★ SLIDES 1+2 -- ABBY

**Abigail:**

**Slide 1:**

“Hello everyone! We’re The Krusty Compost Crew: Today we will be presenting our project Binny & Bloom, and we’re excited to show you our final product:

**SLIDE 2:**

an **Inventory-Powered Recipe Generator, Donation Advisor, and Composting Assistant** — all wrapped in a cute, friendly interface powered by AI.

Our goal was simple:

**Help people reduce food waste using smart inventory tracking and AI-generated suggestions.**

Let’s jump right in.”

## ★ SLIDE 3 - PROBLEM STATEMENT — (Presenter: Taja)

**Start on Slide: 3**

**Taja:**

“Food waste is a massive issue — both environmentally and financially. People buy groceries, forget about them, and end up throwing away perfectly good food.

We wanted to solve that by creating a system that:

- Tracks your inventory
- Warns you when items are about to expire
- Suggests recipes using what you already have
- Helps you donate safe items
- And guides you on composting expired ones

Basically, we wanted to turn your fridge into a sustainability sidekick.”

## ★ SLIDES 4+5. ARCHITECTURE OVERVIEW — (Presenter: Luis)

SLIDE 4:

“Let’s talk architecture.

Our backend is built with **Flask**, and our database is **SQLite**, which stores:

**SLIDE 5:**

- Inventory items
- Players
- Expiry dates
- Sustainability metrics
- Saved recipes

We also have AI integration through a model on OpenRouter, a validator for safe recipes, front-end templates with CSS, and a CLI version for terminal users. “

## ★ SLIDE 6 - ABBY

We also implemented:

- Automatic flagging for raw meat, perishables, donation eligibility, and compostability
- A scoring system that tracks how much value you save by cooking instead of wasting
- Expiry detection that triggers warnings on the dashboard

And all of this connects to our AI layer, which reads the inventory, respects dietary needs, and generates JSON-formatted recipes, donation lists, or composting suggestions.”

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★ **SLIDE 7 AI INTEGRATION — (Presenter: Emma) SLIDE 7**

**Emma:**

“Our AI integration is the heart of the project.

We built a custom wrapper that:

- Sends the user’s inventory to the LLM
- Enforces strict JSON output
- Validates recipes using our own validator
- Checks allergies, dietary needs, expiry status, and unit conversions
- Rejects unsafe or impossible recipes
- And regenerates responses if needed

We also support four AI modes:

1. **Chat**
2. **Recipe generation**
3. **Donation advisor**
4. **Compost advisor**

The validator ensures the AI never invents ingredients, never uses expired items, and always respects the player’s restrictions.”

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## ★ SLIDE 8 – JAY ( Emma)

### **Keeping Recipes Safe & Realistic**

“This slide outlines our Recipe Validator, the engine that ensures every generated recipe is actually feasible and safe.

The logic is built on three pillars:

- Logistics: It verifies stock availability and standardizes all units—grams, milliliters, or counts.
- Safety: It cross-references your personal profile, automatically flagging expired items, allergens, and dietary conflicts.
- Precision: It enforces realistic cooking amounts by matching requirements against your current pantry levels.

Essentially, this system prevents impossible cooking scenarios before the user even starts, turning raw inventory into a guaranteed success.”

## ★ SLIDE 9 – EMMA

### **Your Sustainability Command Center**

“For tracking efficiency in Binny and Bloom, we use an equation to show efficiency score.

It’s going to be the total value of items ate/donated/composted over the total purchase value of items in the inventory, times 100 to get the percentage. Below are the sustainability tiers we have introduced, where a user will be assigned a tier based on their sustainability score. We also have expiry warnings for items that will expire soon, and saved recipes.”

## ★ SLIDE 10 – TAJA

"Next, we tackle waste management with our Expiry Tracking system. We categorize every item into 'Expired,' 'About to Expire,' or 'Fresh' for instant pantry visibility.

To ensure users act in time, the system uses proactive notifications: a 4-day warning for early heads-up and a 1-day urgent alert for immediate attention.

These are reinforced through dashboard flash messages and clear, color-coded visual cues using ⚠️ and 🔔 icons. It transforms a cluttered pantry into an organized, actionable list, ensuring nothing goes to waste. Now, we'll start the live demo."

## ★ SLIDE 11 - LIVE DEMO WALKTHROUGH — (Presenter: Jay) Abby

**Jay:**

"Let's walk through the app from a user's perspective.

### **Step 1: Login**

You log in, and the homepage greets you with your profile picture, score, and navigation menu.

### **Step 2: Inventory**

You can view your items, add new ones, and see expiration dates sorted automatically.

### **Step 3: Dashboard**

Here you'll see:

- Expiring soon warnings
- Urgent 1-day warnings
- Your sustainability tier
- Your efficiency score
- Total value acquired vs saved
- Saved recipes

### **Step 4: AI Menu**

You can ask:

- 'What can I cook?'
- 'What can I donate?'
- 'What should I compost?'

And the AI responds with validated, safe, inventory-accurate suggestions.

### **Step 5: Cooking a recipe**

When you click 'Cook this recipe,' the system:

- Deducts ingredients
- Updates your score
- Increases your total value saved
- And keeps your inventory accurate.”

## **BACK TO SLIDESHOW FOR INDIVIDUAL CONTRIBUTIONS – SLIDE 12**

Abby: Hello, I'm Abigail and for my contribution I focused more on the frontend and overall design of the project. I drew our mascot and helped with both backend and front end (AI side). I helped to implement the LLM into the flask as well as the validator. I also help build the chatbot and load indicator. Aswell i took an active role in the testing of the product and documenting on GitHub.

Taja: My name is Taja, and I created the logic for calculating how many days an item has before it expires, and logic to sort all items in the inventory into three different categories: expired, about to expire, and fresh. I also created the dashboard page and the expiry notification that warns the user twice, once when an item is about to expire in 4 days and one last time when an item is about to expire in 1 day.

Emma: Hi, I'm Emma, and my contributions started with working on the AI integration with our code. I helped find a free LLM for us to use and experimented with system instructions to get it to generate our recipes properly, and I helped work on our recipe validator code. I helped design the sustainability tier system and how our users would be scored, and I did a little work on the dashboard page to show how scores were calculated.

Jay: My name is Jay, my contributions were the parts of the database, gamification for: use, compost, donate, and delete. Unit conversion of items and integration of compost and donation in AI and the integration of dietary needs and allergies in validator along displaying that in the chat box and some designs in the application.

Luis: I'm Luis, and my contributions spanned multiple parts of the project across backend architecture, AI integration, database design, UI/UX, and the CLI version of our system. I worked extensively on the Flask backend, helping structure our routes, database interactions, and overall application flow. I designed and refined the database integration, helping build and debug database queries, fix schema issues, and ensure consistent data flow between the backend and the UI. On the frontend side, I worked on UI improvements, including layout adjustments and integrating dynamic elements like the chat interface and recipe display. I also built the CLI version of the app, including login, inventory viewing, and AI chat features, giving our project a second fully functional interface. Throughout development, I also helped with testing, and ensuring the system behaved consistently across both the web and CLI versions.

## ★ 8. CLOSING AND Q&A — SLIDE 13

**Abigail:**

“Thank you for watching our demo — we’d love to answer any questions!”

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